- It is essential for clotting of blood.
- It regulates the permeability of capillary walls.
- It is essential for the contraction of heart and muscle.
- It regulates the excitability of nerve fibres and nerve centres.
- It acts as an activator for the enzymes present in the gastric juice.
- It plays an important role in maintenance of health.
- Required for proper foetal growth.
- It speeds up all healing process.
- It is essential for proper utilization of phosphorus and vitamins A, C and D

Food Sources of calcium:

The richest source of calcium among animal foods is milk and among vegetables it is green leafy vegetables. Among green leafy vegetables, amaranth, fenugreek and drumstick leaves are particularly rich in calcium. Ragi is the main source of calcium. Sesame seeds with husk and small dried fish are also good sources of calcium.

Symptoms of Calcium deficiency:

- Bone mass is reduced when calcium deposit is less.
- Rickets in children, Osteomalacia in adults, Osteoporosis in old age occurs.
- Decreased growth rate.
- Very often fractures occur due to brittle bones.

11.4.4 Zinc

Zinc is an essential trace element which plays an important role in our body. Our body contains 2-3 grams of zinc. It has been found to be present in the hormone insulin. It plays an essential role in the formation of DNA and RNA. It aids in the healing of burns and wounds.

Functions:

- It plays a vital role in cell division and growth especially during pregnancy and prevents congenital abnormalities and premature delivery.
- It plays an important role in maintaining fertility in males.
- It provides immunity to our body.
- It helps in healing cuts, wounds, acne and rashes.
- It is important for healthy vision and prevents night blindness and cataracts.

Food Sources of Zinc:

Seafoods, meat, eggs are good sources of Zinc. Milk and milk products, whole cereals, pumpkin seeds, cashewnuts, spinach, legumes contain considerable amounts.

Symptoms of Zinc deficiency:

- Stunted growth.
- Loss of appetite.
- Dry and rough skin.
- Dull brittle hair.
- Brittle nails with white spots.
- Loss of memory.
- Reduced sense of taste and smell.
- Delayed healing of wounds.
- Frequent infections and acne.
- Diarrhoea and pneumonia can be fatal.

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